AutoDESN and AutoGP: nonlinear heteroscedastic seasonal dynamics

NHS1 | completed fresh comparison | study initiated 23 October 2026

One new series combines threshold nonlinear dynamics, an asymmetric nonsinusoidal seasonal pattern and persistent, asymmetric conditional variance. No deterministic trend is present. The DGP, seed, prior controls, architecture candidates and forecasting protocol were fixed before fitting. Every model is fitted fresh; no C1 forecast output is reused.

Data-generating process

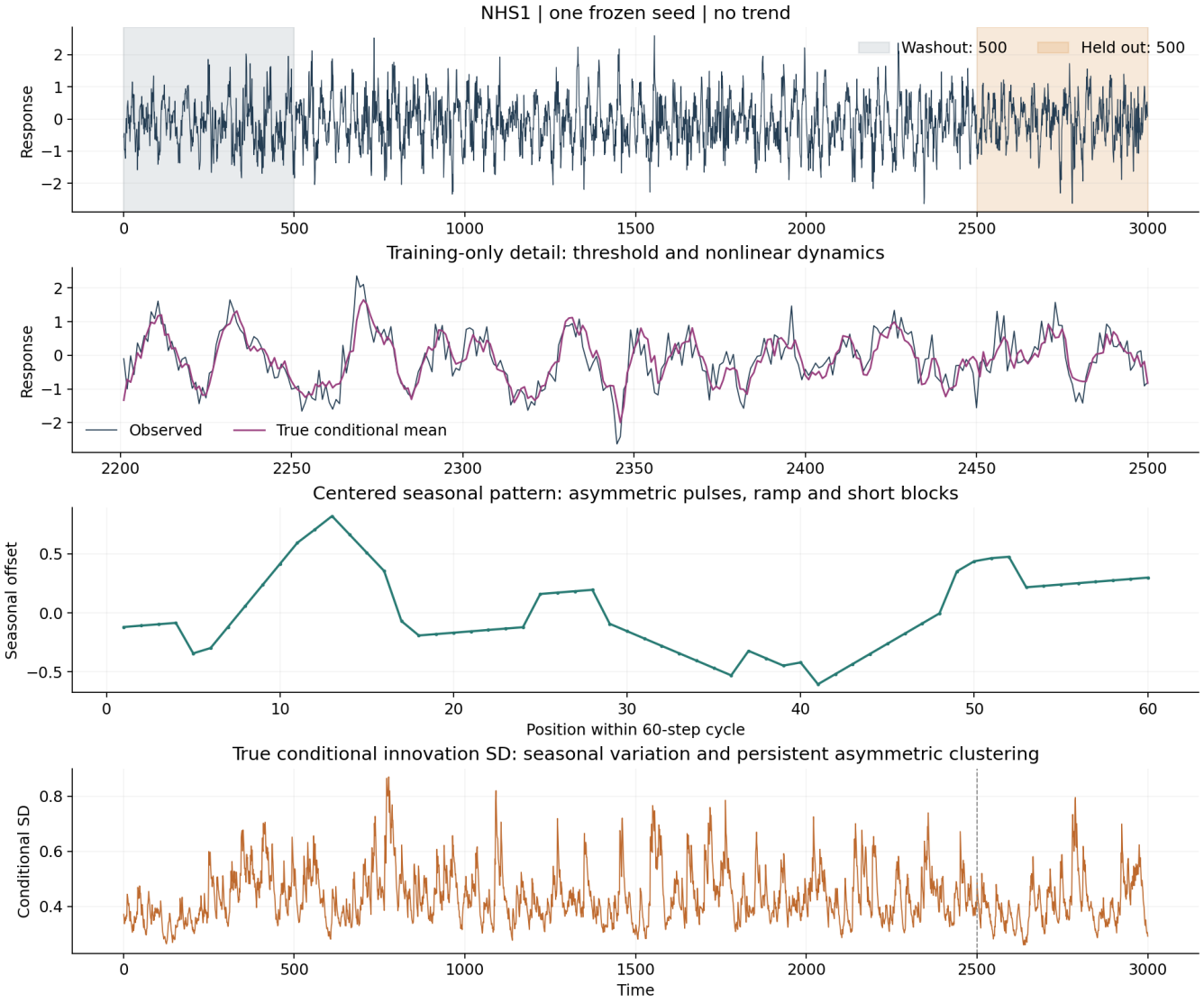
Write y(t)=s(t)+x(t). The centered period-60 seasonal offset combines two asymmetric triangular pulses, a ramp and a 12-step block pattern. The residual conditional mean is a(x1)*x1 +0.28*tanh(1.4*x2) -0.25*tanh(x12) +0.22*tanh(x60) +0.20*tanh(1.5*x1*x2), where xk=x(t-k), a(x)=0.58 for x<0 and 0.22 otherwise.

The innovation is epsilon(t)=sqrt(h(t))*z(t), z(t) standard Gaussian. h(t)=omega(t)+0.06*epsilon(t-1)^2+0.10*epsilon(t-1)^2*I[epsilon(t-1)<0]+0.82*h(t-1). The baseline omega(t) is positive, periodic and dependent on the previous residual; exact pulse definitions are in dgp.R. It is bounded by 0.046 and expected volatility persistence is 0.93. The residual linear coefficient is bounded by 0.58 and other location terms are bounded. The equations specify no trend, and these bounds give bounded second moments. The forcing is periodic; covariance stationarity and ergodicity are not assumed.

Setting	Frozen value
DGP	Seed 1766021; 6,000-step burn-in; 3,000 retained observations
Split and forecast grid	500 washout; 2,000 training; 500 test; 471 origins x 30 leads
Requested normalized RHS	tau0=0.001; slab_scale=1; slab_df=4; c^2~IG(2,2)
Other priors	Unit half-Cauchy global multiplier/local scales; intercept N(0,10 variance); observation variance IG(3,1); ridge precision Gamma(2, rate 2)
Readout basis	Training-only column z-scores on rows 501:2500; no division by sqrt(d)
Primary forecast budget	192 joint posterior draws x 8 recursive innovation paths

All fitted likelihoods retain constant observation-variance parameters. This is a calibration stress test under misspecification, not a claim that their conditional variance models match the DGP. Truth quantities are isolated from the observed-only training interface.

The single frozen realization



The seed was not selected by inspecting forecast errors. DGP truth is used for these plots and post-fit calibration assessment only. Models receive the two-column time/response file. There is no full-series centering, and no held-out response is used to fit a scaler.

Pooled forecast scores

Model	RMSE	MAE	CRPS	NLL
AutoDESN RHS: 0.001 / 1	0.578871	0.447044	0.323882	0.886745
AutoDESN RHS: 1 / 2	0.573141	0.442263	0.320491	0.876026
AutoDESN ridge	0.639339	0.498066	0.358653	0.977519
Native AutoGP	0.666443	0.523538	0.386559	1.103234

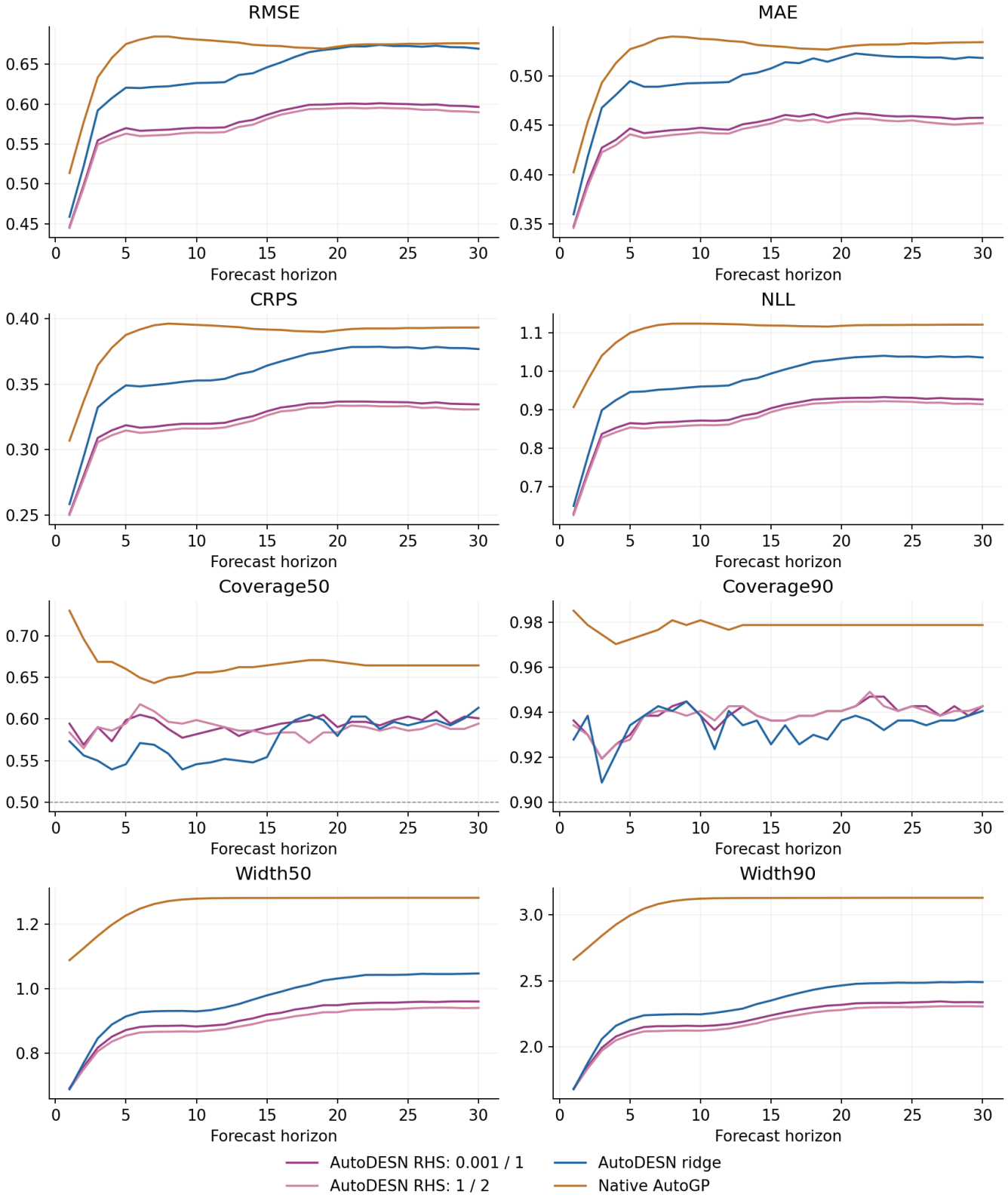
Predictive intervals

Model	Cov.50%	Width50%	Cov.90%	Width90%
AutoDESN RHS: 0.001 / 1	0.5930	0.9067	0.9384	2.2109
AutoDESN RHS: 1 / 2	0.5893	0.8891	0.9382	2.1787
AutoDESN ridge	0.5751	0.9682	0.9338	2.3235
Native AutoGP	0.6657	1.2597	0.9782	3.0726

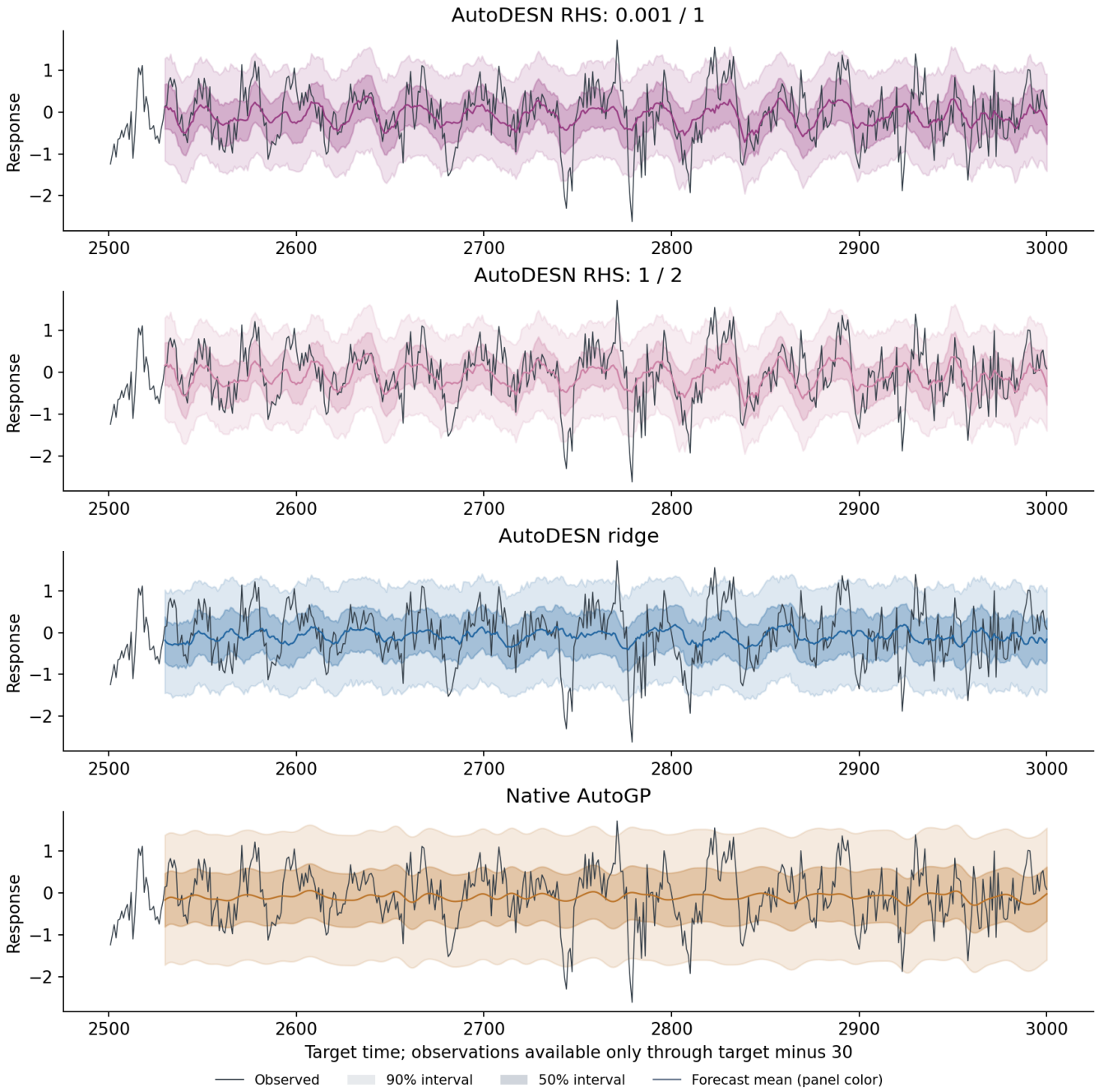
Each model has exactly 14,130 dependent origin/lead pairs. Lower RMSE, MAE, CRPS and negative log predictive density (NLL) are better descriptively. Coverage and interval width must be assessed together. Native AutoGP marginal mixture scores are analytic; AutoDESN CRPS and intervals use recursive draws, and its NLL uses the conditional Gaussian mixture over simulated histories.

One realization cannot establish statistical superiority across DGPs or independent replications. Small gaps must also be assessed against posterior and predictive Monte Carlo error. No hyperparameter search uses these held-out outcomes.

Forecast quality by horizon

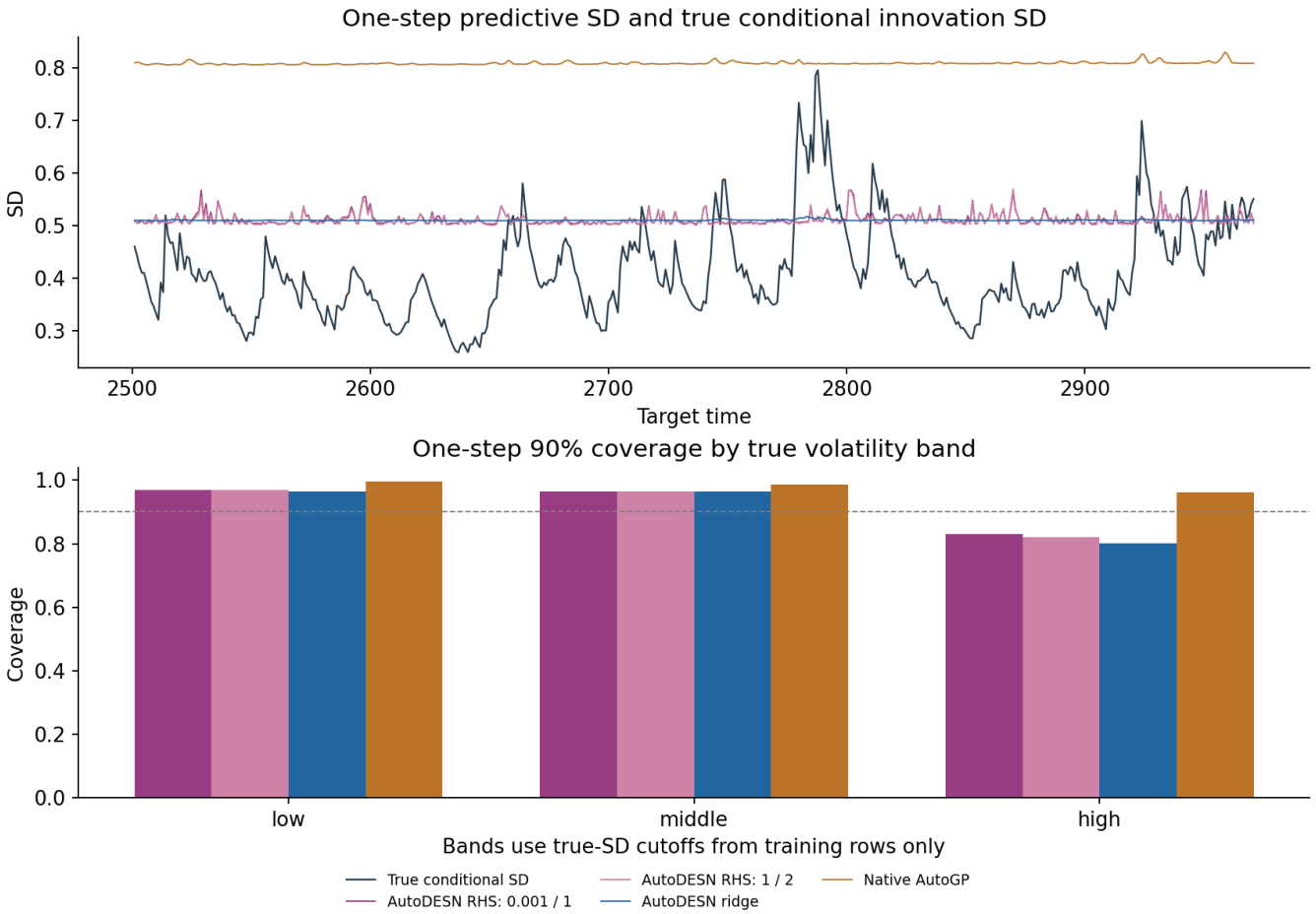


Rolling forecasts and predictive intervals



Horizon 30 at all 471 origins. Parameters and architecture weights are frozen at origin 2500. Observed reservoir states update only between origins. Within a forecast, simulated responses drive all later states; future observations are absent. Native GP conditioning likewise incorporates only observations available at each origin, with kernel parameters and mixture weights fixed.

Calibration under heteroscedasticity

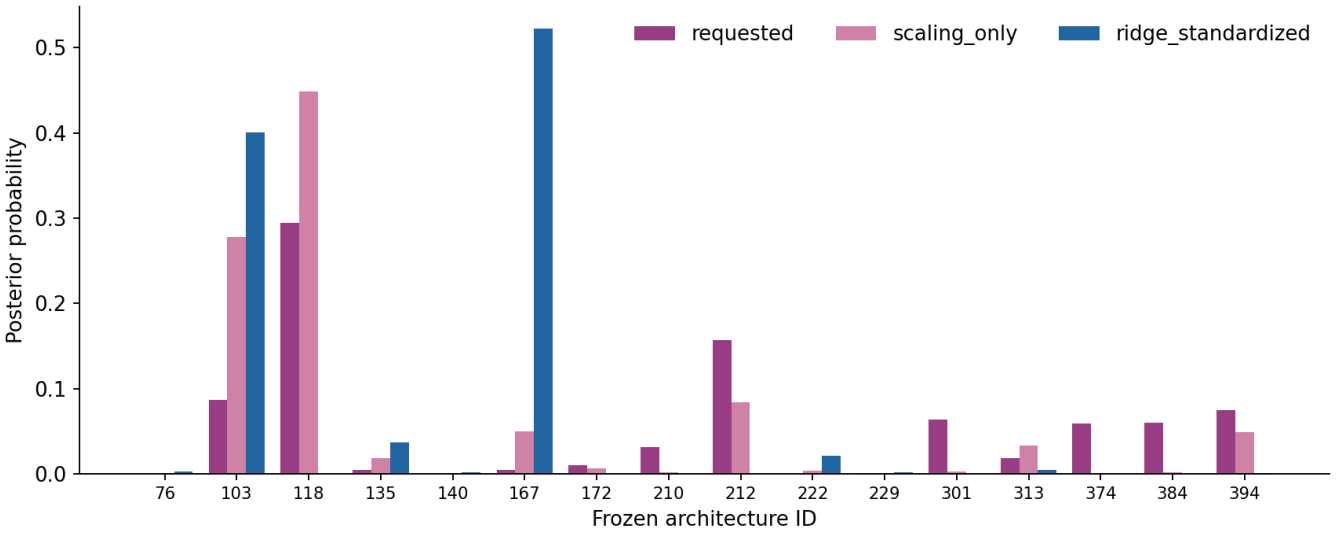


Only horizon 1 is compared with the true conditional innovation SD. A future realized conditional SD would not equal the marginal predictive SD of a multi-step forecast. Low/middle/high bands use the two training-only true-SD tertile cutoffs and are reporting diagnostics, not features supplied to the models.

Model	Band	Pairs	90% coverage	Predictive SD	True SD
AutoDESN RHS: 0.001 / 1	low	222	0.968	0.511	0.342
AutoDESN RHS: 0.001 / 1	middle	143	0.965	0.514	0.420
AutoDESN RHS: 0.001 / 1	high	106	0.830	0.513	0.544
AutoDESN RHS: 1 / 2	low	222	0.968	0.509	0.342
AutoDESN RHS: 1 / 2	middle	143	0.965	0.513	0.420
AutoDESN RHS: 1 / 2	high	106	0.821	0.511	0.544
AutoDESN ridge	low	222	0.964	0.510	0.342
AutoDESN ridge	middle	143	0.965	0.510	0.420
AutoDESN ridge	high	106	0.802	0.512	0.544
Native AutoGP	low	222	0.995	0.808	0.342
Native AutoGP	middle	143	0.986	0.809	0.420
Native AutoGP	high	106	0.962	0.811	0.544

Bayesian architecture averaging and shrinkage

For slopes, beta_j | tau,lambda_j,c² ~ N(0,v_j), v_j=c²*tau²*lambda_j²/(c²+tau²*lambda_j²), independently of observation variance sigma². The global multiplier u and local lambda_j have unit half-Cauchy priors; tau=tau0*u. The intercept has variance 10. Architecture moves integrate every coefficient using C_m=sigma²*I+X_m*diag(10,v)*X_mᵀ, retaining its determinant and normalized Gaussian density. This is the normalized RHS, not the augmented rhs_ns prior.



Scenario	D	Layer widths	Input lags	Layer leaks	Layer radii	Seed
requested	1	50	60	0.6	0.9	29
scaling_only	1	50	60	0.6	0.9	29
ridge_standardized	1	25	5	0.15	0.1	47

Leading architectures are identified by the posterior mass, without changing the predictive mixture. Full candidate definitions, all 426 architecture probabilities and numerical uncertainty summaries are archived.

RHS scales and shrinkage

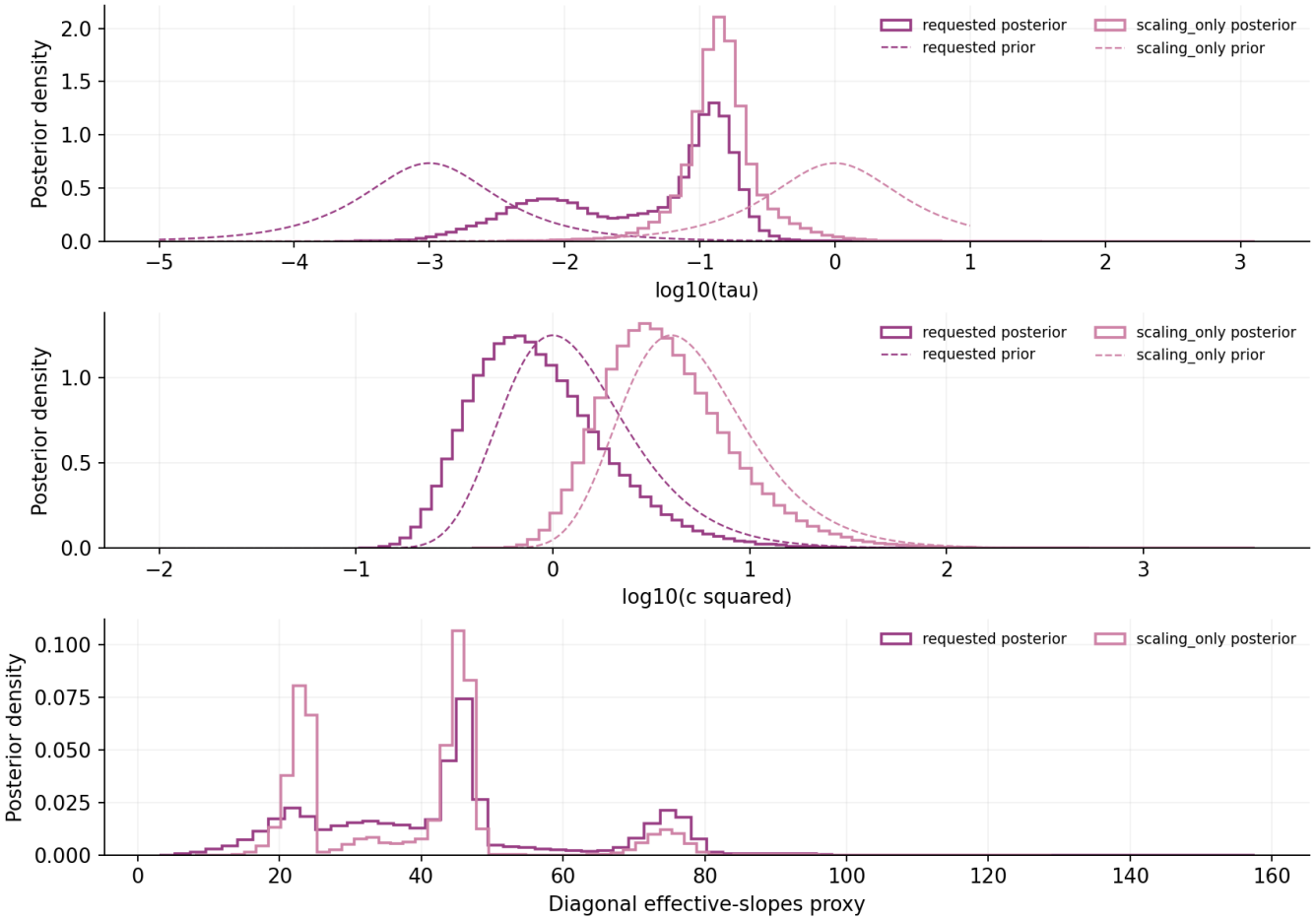
Scenario	Leading candidate	Mass (indicator SE)	tau median [5%,95%]	c^2 median [5%,95%]
requested	118: broad_046	0.294 (0.018)	0.07463 [0.002663, 0.2103]	0.7817 [0.2819, 3.694]
scaling_only	118: broad_046	0.448 (0.018)	0.1404 [0.05224, 0.4235]	3.537 [1.313, 15.85]
ridge_standardized	167: broad_095	0.522 (0.004)	-	-

The indicator-based SE uses the leading model frequency ESS as an approximate sampling-error diagnostic; conditional probability mass estimates have their own batch MCSE in the archived tables. Neither estimates variation across independent time-series realizations.

requested diagonal effective-slopes proxy: median 44.196, 5%-95% [17.538, 77.011]. This sums d_j*(X'X)_jj / (sigma²+d_j*(X'X)_jj), using slope prior variances d_j; it ignores feature correlations and is not a count of selected coefficients.

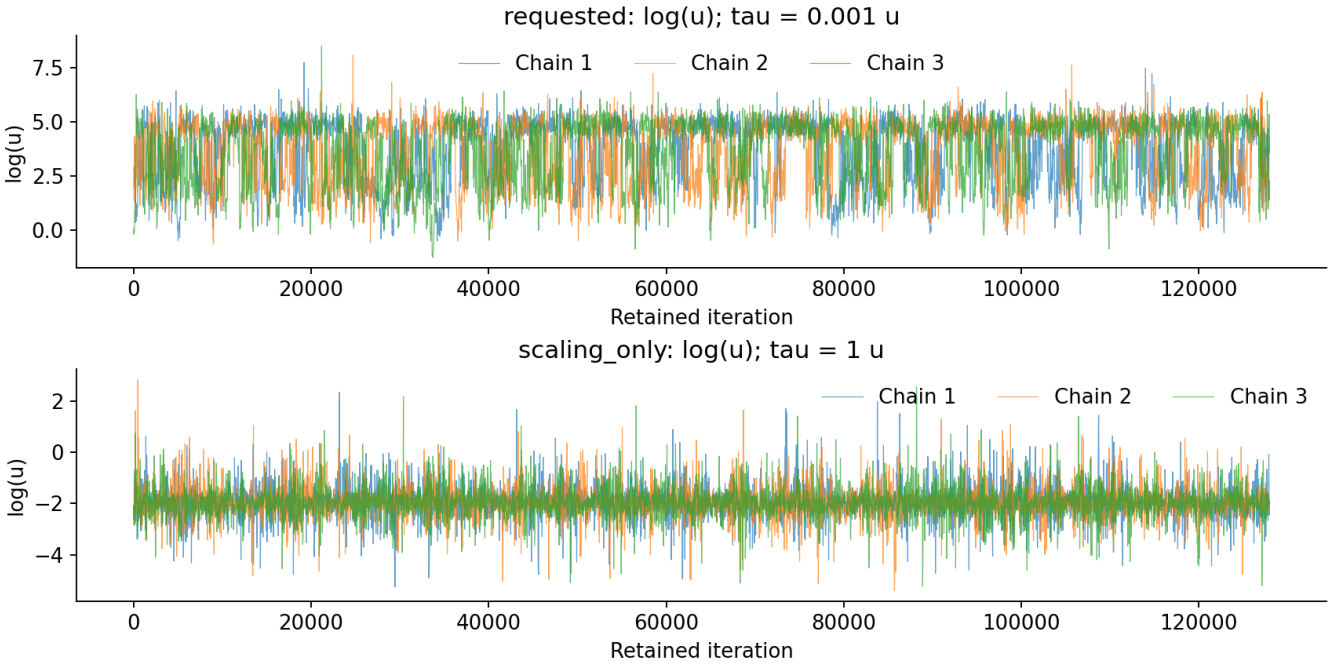
scaling_only diagonal effective-slopes proxy: median 43.450, 5%-95% [20.790, 74.117]. This sums d_j*(X'X)_jj / (sigma²+d_j*(X'X)_jj), using slope prior variances d_j; it ignores feature correlations and is not a count of selected coefficients.

tau0 is a prior multiplier, not a fixed posterior tau. Global, local and slab scales are sampled. The normalized Gaussian slope density and every architecture-dependent Gaussian determinant are retained. Layerwise common auxiliary coordinates and product-preserving interweaving preserve the same normalized RHS target; inactive proper priors integrate to one. No test-error screening substitutes for architecture averaging.



Solid histograms are posterior densities; dashed curves are the normalized half-Cauchy global-scale and inverse-gamma slab priors after the log10 coordinate transformation. The effective-slopes panel is an approximation, with no implied variable-selection threshold.

Retained-chain diagnostics



Scenario	Max rank R-hat	Min bulk ESS	Min tail/frequency ESS	Kept/chain
requested	1.00992	499.6	627.3	128000
scaling_only	1.00391	796.7	796.7	128000
ridge_standardized	1.00040	3481.6	5608.3	6000

Three chains; continuous split rank/folded R-hat<=1.01 and bulk/tail ESS>=400. Six fixed training-row fitted means are additionally monitored across architectures with different coefficient dimensions. The mean_log_local monitor averages the common auxiliary vector, including inactive prior coordinates; it is not an active-coefficient shrinkage summary. Architecture indicators with posterior mass>=1% also pass rank R-hat/frequency ESS, plus the R spectral-ESS and absolute chain-frequency-range gates. Binary frequency ESS is shown in both ESS columns. Checks pass before forecasting. These monitored diagnostics do not prove convergence or accurate rare-model probabilities. Full traces, controls, chain exploration and all failed budgets are archived.

Architecture exploration

Scenario	Models visited (chains 1/2/3)	Retained switches (chains 1/2/3)
requested	72 / 74 / 77	7696 / 8808 / 8491
scaling_only	66 / 57 / 59	10007 / 9063 / 9122
ridge_standardized	19 / 21 / 19	3091 / 3239 / 3160

Visits and switches describe exploration; they are not convergence criteria. Proposal/acceptance counts include burn-in and are retained with the exact controls. Architecture indicator diagnostics concern the retained chains.

Sampling and predictive budgets

Scenario	Burn/chain	Kept/chain	Max R-hat	Min ESS	Max mass range	Pass
requested	4000	16000	1.08185	146.5	0.352	False
requested	8000	32000	1.01575	180.6	0.145	False
requested	12000	64000	1.01151	404.6	0.040	False
requested	16000	128000	1.00847	698.9	0.040	True
scaling_only	16000	128000	1.00313	1081.6	0.044	True
ridge_standardized	2000	6000	1.00037	3571.9	0.009	True
requested_initial_proposal	2000	6000	1.25656	91.6	0.494	False
requested_initial_proposal	4000	16000	1.05740	146.7	0.121	False
requested_initial_proposal	6000	32000	1.02389	170.3	0.137	False
requested_initial_proposal	10000	64000	1.36972	342.8	0.819	False
requested_rebalanced_64k	10000	64000	1.10079	162.6	0.190	False
scaling_only_joint_initial	4000	16000	1.01316	135.1	0.055	False
scaling_only_joint_initial	8000	32000	1.01865	246.9	0.065	False
scaling_only_joint_initial	12000	64000	1.01342	401.2	0.178	False
scaling_only_joint_initial	16000	128000	1.00404	758.0	0.098	False

The displayed R-hat/ESS columns concern continuous parameters. Pass additionally requires architecture diagnostics. After two failed model-only proposals, a joint architecture/log-scale Student-t mixture is fitted to completed training chains and frozen before fresh production. Four original model-only moves and two joint moves per sweep retain exact Gaussian coefficient-collapsed targets, prior/Jacobian constants and Hastings corrections. Initial production budgets are 16k/32k/64k/128k kept per chain. After the scaling-only final budget failed the 5-point architecture-frequency-range gate, a same-prior proposal recovery declares one fresh 3 x 128k budget. Shared training is reported once below. No failed screen is waived. No test scores inform recovery.

Predictive Monte Carlo sensitivity

Scenario	N x K	RMSE (3 origins)	CRPS	NLL
requested	192 x8	0.55961	0.31709	0.86781
requested	384 x32	0.55845	0.31529	0.86568
scaling_only	192 x8	0.55083	0.31269	0.85162
scaling_only	384 x32	0.55550	0.31366	0.85823
ridge_standardized	192 x8	0.62933	0.35598	0.97853
ridge_standardized	384 x32	0.63021	0.35517	0.97915

All 24 origin/budget combinations per model are archived: origins 2500, 2735, 2970; N=192/384 and K=4/8/16/32. Each of the eight N x K settings is aggregated across three origins, giving 90 dependent pairs rather than 14,130. The same qualified posterior is reused. Native GP additionally exports full run-by-run forecast sensitivity for its two independent eight-particle SMC fits.

Native AutoGP and finite-particle sensitivity

Fresh authors' AutoGP 0.1.19 at revision 2ad372db2b8bea176e3b32a769bfb5d6cff80866, Julia 1.10.10. Native grammar/priors, depth 4, two independent eight-particle SMC fits; unchanged stage-wise schedule and HMC controls. Native save/reload and three-origin Gaussian-conditioning checks passed. These verify transport, not SMC posterior exploration.

Run	Particles	Final particle ESS	Distinct kernel states	Largest weight
1	8	5.459	8	0.2649
2	8	4.241	8	0.3812

Distinct exported kernel states include continuous parameter values, not just tree structures. Particle ESS can be reset by resampling; it does not establish exploration of kernel space. Run-by-run forecast sensitivity measures variation between two finite-particle approximations, not independent DGP replications.

SMC run	Pairs	RMSE	CRPS	NLL
1	14130	0.669804	0.386910	1.100006
2	14130	0.671053	0.389448	1.109938

Training stage	Structural MCMC steps	HMC steps	HMC step size
32	40	4	0.0200000
64	40	4	0.0100000
128	40	4	0.0050000
256	30	3	0.0025000
512	20	3	0.0012500
1000	15	2	0.0006400
1500	10	2	0.0004267
2000	10	2	0.0003200

Each run has eight particles and maximum grammar depth 4. Leapfrog steps: 5; HMC exit control: 2; step size=0.02*32/n; seeds: 1166020 and 1166021. These are the original frozen C1 controls applied to the new data. The two-run predictive mixture uses the fitted particle weights; model weights remain fixed between origins.

Measured runtime and memory

Shared raw builder: 174.04s wall; frozen standardized-design preparation: 18.19s. The complete bank-building command, including startup and serialization, took 219.00s with parent peak RSS 5.01GiB. Reservoir candidates and state timing match across all AutoDESN priors.

Scenario	Pilots(s)	Completed chain(s)	Observed fit wall(s)	Forecast compute(s)	State update(s)	Observed peak GiB
requested	0.00	3768.36	4003.91	96.51	3.82	7.57
scaling_only	0.00	1573.81	5795.43	65.39	2.88	6.81
ridge_standardized	0.00	70.34	101.16	31.92	0.77	3.35

Additional failed training cost (requested_initial_proposal): 1438.78s completed inference plus 29.19s pilots. These training fits issued no forecasts. The requested-prior source is shared across RHS cases; the scaling-only joint source is additional recovery cost for that control.

Additional failed training cost (requested_rebalanced_64k): 637.31s completed inference plus 0.00s pilots. These training fits issued no forecasts. The requested-prior source is shared across RHS cases; the scaling-only joint source is additional recovery cost for that control.

Additional failed training cost (scaling_only_joint_initial): 3305.98s completed inference plus 0.00s pilots. These training fits issued no forecasts. The requested-prior source is shared across RHS cases; the scaling-only joint source is additional recovery cost for that control.

Execution interruption (scaling_only): last observed invocation wall 2725.46s. Its completed chains are counted once in chain compute; the unfinished chain was recomputed. The first interrupted invocation's stdout was overwritten before backup; its last timing observation and resume logs are retained. Fit wall sums recorded invocations, including this observation, and includes repeated cache loading and diagnostics.

Native AutoGP summed SMC callback training: 2978.81s. Native invocation including startup, training, export and conditioning contracts: 3115.11s. Rolling conditioning/scoring/export: 21.95s. Native-process peak RSS: 1.931GiB. Timer scopes differ from AutoDESN per-origin compute; no clean end-to-end speedup claim follows.

Inference totals include completed failed budgets; interrupted partial work is retained separately in wall-time logs. Forecast compute sums per-origin calls and excludes state preparation, scoring/export and sensitivity. Parent VmHWM excludes forked workers; Native Julia memory uses Linux wait4 child rusage. R uses one BLAS thread; the original native AutoGP script uses two BLAS threads and one Julia thread. Large inference/forecast jobs run sequentially. Hardware quota and memory limits are archived. Light validation/integrity checks and initial dependency setup may overlap inference; these are observed runtimes, not isolated hardware benchmarks.

Validation, limits and reproducibility

All 426 admitted architectures share the raw bank and frozen feature transforms. Maximum absolute training column mean 8.77e-14; maximum sample-SD error 2.22e-16. The API freezes exclusions for near-constant features; the intercept has its own proper prior and raw-coordinate transport includes its centering shift.

Fresh checks for the changed production source passed 652 focused assertions (one loaded-DLL context skip covered by 76 source-only checks), 77 R study contracts, 6 Python scoring/causality contracts, 4 independent diagnostic tests and 32 independent posterior-reference checks. Fresh NHS1 tests passed 12 DGP assertions, 24 workflow assertions and 6 dataset/scoring/conditioning contracts. The optional joint sampler passed 6 independent normalized-density states, 36 ordered acceptance ratios and 4 posterior functionals against independent integration; disabling it preserves previous draws exactly. Native GP transport/conditioning contracts and R/C++ small-chain parity are separately checked. These checks do not establish full-size posterior convergence: all production chains must pass their own parameter/architecture and independent rank/ESS screens before forecasting. Five further learned-local/interwoven joint-kernel posterior functionals agree with independent four-dimensional normalized-prior integration (three chains; inactive local priors integrated exactly).

Fresh short computational benchmark: median R 1.141s versus native kernels 0.675s (1.69x). Four standardized synthetic designs, 200 rows, up to 64 slopes, 100 burn +800 retained draws, three timing repetitions; setup/compilation excluded. Maximum coupled-draw discrepancy 4.96e-12; identical model paths. Joint proposals are disabled in this short kernel benchmark; production timings include them. This is a computational check, not evidence of posterior convergence or a full-study speedup.

Independent path reconstruction verifies scores at the first, middle and last origins. Future-data perturbation changes every unavailable response by 100 while holding posterior draws and innovations fixed; forecasts remain identical. Archived joint checkpoints are replayed against saved paths. The numeric archive has part-level, complete-archive and internal SHA256 checks, plus a restored-analysis round trip.

The frozen reservoir grid includes input lag 60, matching the DGP season and lag 60 term. Broader stress studies should vary periods and independent seeds. Gaussian conditional variances are deliberately misspecified for this heteroscedastic DGP. Poor calibration in high-volatility periods motivates a future variance-model extension; such an extension would be a distinct benchmark with its own prior and validation. A single series and overlapping origin/lead pairs do not establish general superiority. Preprocessing is treated as frozen rather than jointly inferred.

Release remains draft. The inherited scoped repository R CMD check has one nonportable-path error, three documentation/non-ASCII warnings and two notes. Focused tests and numerical checks are reported separately; no clean package-wide release check is claimed.

See README.md and RESULTS.md in validation/autodesn/nonlinear_heteroscedastic. generate.R freezes the DGP; run.R/fit_joint.R/execute_joint.py fit the three AutoDESN cases; run_gp.py fits pinned Native AutoGP; export.R, archive.R, replay.R, analysis.py and report.py reproduce evidence and reporting. Existing C1 artifacts are preserved. Draft PR: https://github.com/AntonioAPDL/exdqlm/pull/4.